

AddLyft · Social Post Workflow Audit

Complete Text + Image pipeline overview for client review
Vertex AI · Gemini text + Nano Banana image · Local R&D showcase
Report date: 23 Sep 2026 · Scope: social-post only (video / Veo excluded)

How to read this report. Every cost and timing number is tagged: **Measured** = taken from this project's usage logs; **Configured estimate** = calculated with the rates coded in the app (not a GCP invoice); **Unknown** = not recorded or not determined from the current implementation. Nothing here invents production SLAs or billing totals.

1. Executive summary

The Social Post Workflow turns a short advertising brief into two reviewable drafts in one screen: (1) structured social copy (caption, offer, CTA, hashtags) and (2) an optional portrait poster image, optionally guided by up to three reference photos. A fixed demo store identity (**AMPM Woodstock**) is applied automatically so the brief does not need to repeat the store name. Outputs are labeled AI drafts — the system does not publish to social networks.

On a successful “Generate post + poster” click, the UI calls two backend APIs **in parallel**: one Vertex text call and one Vertex image call (happy path = **2 model calls**). Regenerate updates text only (1 model call). Image generation dominates both time and cost.

Field	Value
Primary UI	http://127.0.0.1:5173/ (React)
Unified API	http://127.0.0.1:8000 (FastAPI)
Text model	gemini-2.5-flash-lite (Vertex)
Image model	gemini-3.1-flash-lite-image (Vertex, 9:16, 1K)
Typical combined cost (Configured estimate from Measured logs)	≈ \$0.041 text + ≈ \$0.0344 image
Typical image latency (Measured)	≈ 8–11.5 seconds wall time (recent log samples)
Text latency (Measured)	Unknown — text usage log does not record wall_seconds

2. Architecture and generation flow

User enters a brief (and optionally 0–3 reference images) in the React UI. Vite proxies `/api` to the FastAPI backend on port 8000. The backend owns both text generation and image generation (image logic reused from `social-post-image/`). A standalone image HTML page on port 8787 remains optional for debugging only.

2.1 End-to-end steps (Generate)

- 1 Validate brief length / blocklist; apply selected store context (AMPM Woodstock).
- 2 Start text + image requests together (parallel).
- 3 Text: Gemini Flash-Lite returns JSON matching caption / offer / CTA / hashtags; output shape and claim checks run before response.
- 4 Image: build role-aware poster prompt; resize refs; one Nano Banana image call; save poster + meta under `social-post-image/outputs/`.
- 5 UI shows whichever side succeeded; errors are shown separately (partial failure OK).
- 6 Compact usage strip: tokens / estimated cost (and time for image).

API call count (happy path). Generate = 2 Vertex calls. Regenerate (text variation) = 1 Vertex text call. Text may retry once on non-safety failures (up to 2 text calls if transient error). Normal image generate is exactly one model call — no automatic edit pass.

3. Models and their roles

Pipeline	Model ID	Role	Source
Text	gemini-2.5-flash-lite	Structured social copy (JSON)	Configured default (GEMINI_MODEL)
Image	gemini-3.1-flash-lite-image	9:16 poster (1K)	Configured default (IMAGE_MODEL)

Text uses temperature 0.7 on first generate and 1.05 on regenerate (configured in code). Image aspect 9:16 and size 1K are configured defaults; the lite image model does not support 2K in this setup.

4. APIs and interfaces

Field	Value
GET /api/health	Service health + text model + store brand
POST /api/generate	JSON brief → caption, offer, CTA, hashtags + usage
POST /api/generate-image	Multipart prompt + refs → base64 poster + usage
Optional SPI UI	Port 8787 — image-only HTML (not required for showcase)

5. Tokens, latency, and cost

Costs below are **Configured estimates** using rates stored in the project (and matching root .env). They are not Google Cloud invoice lines. Token counts and image wall times are **Measured** from local JSONL logs.

5.1 Unit rates used by the estimator

Pipeline	Token / unit type	USD per 1M tokens	Kind
Text	Input	\$0.10	Configured estimate
Text	Output (+ thoughts)	\$0.40	Configured estimate
Image	Input	\$0.25	Configured estimate
Image	Text output	\$1.50	Configured estimate
Image	Image output	\$30.00	Configured estimate

Image size fallback table (when modality metadata is thin): 1K ≈ 1120 image output tokens (configured in image_usage.py).

5.2 Measured text samples (logs/usage.jsonl)

When (UTC)	Source	Tokens (in/out/total)	Est. USD	Wall time
2026-09-22 19:41	api_generate	367 / 156 / 523	\$0.000099	Unknown
2026-09-22 19:33	api_generate	367 / 133 / 500	\$0.000090	Unknown
2026-09-13 13:54	test_vertex_call	47 / 82 / 129	\$0.000038	Unknown

Model on these rows: gemini-2.5-flash-lite. Text wall_seconds is not logged → latency Unknown for text.

5.3 Measured image samples (social-post-image/logs/image_usage.jsonl)

When (UTC)	Source	Total tokens	Est. USD	Wall (s)
2026-09-22 19:41	unified_ui_generate	4417	\$0.03442	11.43
2026-09-21 18:40	ui_generate	4360	\$0.03441	11.36
2026-09-21 18:37	ui_generate	4360	\$0.03441	11.07
2026-09-21 18:31	ui_generate	1716	\$0.03375	8.04

Model on these rows: gemini-3.1-flash-lite-image · aspect 9:16 · size 1K. Older log lines may show historical Pro / 2K / 16:9 runs — those are not current defaults.

5.4 What a full Generate typically costs

Adding a recent Measured text estimate (~\$0.0001) to a recent Measured image estimate (~\$0.0344) gives roughly **\$0.0345 per full generate** (Configured estimate from Measured token logs). Image is >99% of that cost. Actual cloud bill may differ with promotions, discounts, or rate changes.

6. Input / output and processing

6.1 Text pipeline

In: freeform brief (max 2000 characters) + store_brand (fixed AMPM Woodstock in the demo UI). **Out:** caption, offer, CTA, 3–6 hashtags, usage. Processing: system “facts only” instructions, JSON schema enforcement, Vertex safety filters, then local checks (lengths, hashtag format, light claim grounding for \$ / % style numbers).

6.2 Image pipeline

In: same brief + optional 0–3 reference images + per-image role (product / style / brand / background / auto) + allow-people flag + store. **Out:** poster image (base64), filename, usage (including wall time), person_generation mode used. Processing: build role-aware prompt, resize refs to max side 1600 JPEG, one image generate call, write outputs + meta JSON.

7. Reference-image handling

Field	Value
Maximum references	3 (enforced in UI and API)
Optional?	Yes — zero refs still allowed
Roles	product, brand, background, style, auto (default)
Prep	Convert/resize to RGB JPEG, max side 1600px, quality 90
Allowed types	jpg, jpeg, png, webp, heic, heif
People / faces	Default ALLOW_NONE; user must opt in to ALLOW_ADULT (no silent auto-upgrade)

8. Guardrails and validation

Guardrails sit before and after the models — not only as polite prompt wording.

- **Input:** empty rejection, 2000-char cap, light abuse blocklist (e.g. fake reviews, weapons for sale), file type / ref count checks.
- **Vertex safety filters:** harassment, hate, sexually explicit, dangerous at BLOCK_MEDIUM_AND_ABOVE (configured).
- **Facts-only prompts:** do not invent prices, discounts, dates, or store details; selected store is injected as known context.
- **Text output checks:** required fields, length caps, 3–6 #hashtags, claim tokens (\$ / % / “off”) must appear in the brief.
- **Image people policy:** no faces by default; explicit opt-in required.
- **Human review:** UI labels drafts “not published” with a short verify checklist.

9. Limitations, assumptions, and considerations

- Local R&D / client showcase — not a production publishing platform (no auth, tenant quotas, or Meta/TikTok post APIs).
- Store is a fixed demo default, standing in for a future “select store” step.
- Cost figures are estimator rates × logged tokens — not GCP Console invoices.
- No automatic OCR check that poster text matches the brief exactly.
- Text latency not recorded in usage.jsonl (Unknown).
- Image output token counts may use a size table / modality heuristic when provider metadata is incomplete.
- Claim grounding is lightweight regex — not full legal claim substantiation (advertiser still responsible under truth-in-advertising rules).
- Prior image MVP audit PDF and social-post token report remain useful background; this document reflects the unified Text+Image workflow as of 23 Sep 2026.

10. Source materials used for this audit

Code: backend/, frontend/, social-post-image/, usage.py, social-post-image/image_usage.py. Measured logs: logs/usage.jsonl, social-post-image/logs/image_usage.jsonl. Prior docs: reports/social_post_generation_report.pdf, social-post-image/reports/AddLyft_Social_Post_Image_MVP_Audit.pdf, reports/AddLyft_GCP_Vertex_AI_RnD_Technical_Reference.pdf (context only; video sections out of scope).